

various fronts. This article covers how electronics has been helping in designing and modifying new equipment in order to tackle the Covid-19 pandemic.

Role of electronics

The development and implementation of different technological solutions aimed at combating Covid-19 are rapidly taking shape around the world. There's a wide range of electronic devices that are being used independently or in association with other equipment in the fight against coronavirus like temperature sensors, biosensors, and ventilation machines.

Engineers are taking advantage of modular electronic design, 3D printing, and various other skills to create prototypes of devices that could be critical in the fight against Covid-19. For example, 3D printing can be used to help produce vital hospital supplies. 3D-printing design engineers and specialists are already assisting in the production of respirators, valves, masks, etc. Here are some of the initiatives well supported by electronics towards tackling Covid-19.

Digital initiatives. Digital platforms can help in containing the crisis and save lives. The world needs to embrace other forms of digital technologies as well that will impact livelihood, accelerate various sectors, and strengthen the supply chain and cybersecurity. A coordinated global response using digital technologies is needed to fight Covid-19, with greater impetus on strengthening IT and cybersecurity. For instance, there is a need to enable non-personal data exchange in a secure manner and promote the use of digital solutions in healthcare.

Information provider. The pandemic has highlighted how communication and information dissemination, which promotes accuracy and credibility while preventing the spread of misinformation, is the next level of collaboration in governance. Electronic communication media has played a great role in creating awareness on subjects related to Covid-19, as well as in countering the misinformation related to Covid-19, which otherwise might have created panic in masses.

Different electronic media like television, radio and smartphones show



Thermal cameras for temperature screening (Credit: <https://assets.crowncommercial.gov.uk/wp-content>)

scientific reports, involve informed people in discussions to counter the spread of misinformation, and highlight the importance of precautions to be taken in the wake of coronavirus outbreak. The electronic media through video conferences and online learning has played an important role during the pandemic time.

Electronic entertainment. Even in today's highly advanced and medically adept society, suppressing the virus has required extreme measures of isolation and social distancing on a scale never seen before. It's not yet clear how long the current situation would prevail, and in such times of social distancing, electronic entertainment is the only source of relaxation for masses.

Electronic gadgets such as laptops and phones have greatly helped people to stay in contact with their loved ones.

Electronic assisted medical equipment. Teams of engineers, physicians, computer scientists, and others all over the world are working to develop and implement safe and inexpensive alternative medical equipment that could be built quickly around the world. When it comes to the development of low-cost critical medical equipment like ventilators, personnel protection equipment (PPE), virus-

free masks, etc, electronics plays a great role.

Temperature recording. There are a number of monitoring devices being used to detect Covid-19 by recording the temperature of the general public. Regardless of the level of technology, all such devices make use of remote temperature detection method to measure the temperature of the person/a group of people with the use of infrared waves.

Infrared wave sensors are ideal as these are non-invasive, relatively inexpensive, and can be used from a distance. Internal calibrations from the detector convert the concentration of infrared waves into electricity via a thermopile, thus giving temperature as an output.

Other advanced devices such as thermal imaging cameras detect the temperature by recognising and capturing different levels of infrared light. This light is invisible to the naked eye, but can be felt as heat if the intensity is high enough.

While thermal imaging cameras are generally more accurate, other basic hand-held sensors and monitoring devices are often more than sufficient for recording peoples' temperature.

Advanced monitoring techniques. Other emerging technologies such as